

Performance Testing

Date	4 October 2026
Team ID	6ab2951f1266c775575a6b80
Project Name	Farm Management System
Maximum Marks	5 Marks

Performance Testing

Performance testing is done to evaluate how well the system performs under normal and heavy load conditions. It helps to find response time, throughput and resource usage, and ensures the system is stable, scalable and efficient.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter (or) Apache Benchmark
Type of Testing	Load Testing, Stress Testing
Target Modules / APIs	Login, Add Crop, View Farm, Update Data
Test Environment	Local Server (Development Environment)
Test Date	4 October 2026

Step 2: Test Execution

S.No	Test Scenario	No. of Virtual Users	Duration (min)	Expected Response Time (ms)	Actual Response Time (ms)	Status
1	Login	50	5	< 2000	1450	Pass
2	Add Crop	50	5	< 2500	1780	Pass
3	View Farm Details	100	5	< 2000	1620	Pass
4	Update Crop Data	100	5	< 2500	2100	Pass
5	Logout	50	5	< 1500	1200	Pass

S.No	Test Scenario	No. of Virtual Users	Duration (min)	Expected Response Time (ms)	Actual Response Time (ms)	Status
6	Generate Reports	100	5	< 3000	2450	Pass
7	Search / Filter	100	5	< 2000	1680	Pass
8	Data Export (CSV)	50	5	< 4000	3320	Pass

Step 3: Analysis & Conclusion

Performance Observations	Conclusion
- The system handled up to 100 virtual users without failure. - Average response time remained within the expected limit. - No major errors or timeouts were observed during testing. - Memory and CPU usage were within acceptable range.	The Farm Management System performs well under normal and high load conditions. The response time, throughput and resource usage are satisfactory. The system is stable, scalable and ready for deployment.

Step 4: Recommendations

- Monitor server resources during real-time usage.
- Optimize database queries for faster data retrieval.
- Consider load balancing for higher user concurrency in the future.